

Welcome to CH 221 Organic Chemistry 1!

Dr. Cassie Lilly

Additional Problems CH 1: 43, 46, 48-51, 53, 55, 57-59, 62, 63, 65-68

Goals For Studying Organic Chemistry

- Understand the structure and bonding of organic chemicals
- Learn to correlate the structure with the physical and chemical properties of organic chemicals
- Learn to predict stability and reactivity of molecules
- Learn to speak and understand organic chemistry
- Learn about the reactions that are important in biochemistry
 - All biochemistry is organic chemistry performed by living organisms

Chapter 1

Review from General Chemistry

- Lewis Structures
- Electronegativity
- Polarity
- Formal Charge
- Resonance
- Acid/Base

Orbital Blocks of the Periodic Table

Groups																		18					
1A																		8A					
1	2																	13	14	15	16	17	2
1	2																	3A	4A	5A	6A	7A	He
1	2																	5	6	7	8	9	10
1	2																	B	C	N	O	F	Ne
1	2																	2s ² 2p ¹	2s ² 2p ²	2s ² 2p ³	2s ² 2p ⁴	2s ² 2p ⁵	2s ² 2p ⁶
3	12	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18						
3	12	3B	4B	5B	6B	7B	8B		1B	2B	Al	Si	P	S	Cl	Ar							
3	12	3B	4B	5B	6B	7B	8B		1B	2B	3s ² 3p ¹	3s ² 3p ²	3s ² 3p ³	3s ² 3p ⁴	3s ² 3p ⁵	3s ² 3p ⁶							
4	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36						
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4																							

The Lewis Model of Covalent Chemical Bonding

- In 1916 G. N. Lewis proposed that atoms bond in order to achieve a more stable electron configuration
- Maximum stability results when an atom is isoelectronic with a noble gas
- An electron pair that is **shared** between two atoms constitutes a **covalent bond**

Covalent Bonding in H₂

- Two hydrogen atoms, each with 1 electron $\text{H}\cdot \quad \cdot\text{H}$
- Can share those electrons in a covalent bond $\text{H}:\text{H}$
- Sharing the electron pair gives each hydrogen an electron configuration analogous to helium
- Forming bonds gives off energy
 - Exothermic
- Breaking bonds requires energy
 - Endothermic

Covalent Bonding in F₂

- Two fluorine atoms, each with 7 valence electrons



- Can each share one electron to form a covalent bond



- Sharing the electron pair gives each fluorine an electron configuration like neon

The Octet Rule

- Hydrogen can only have two electrons around it, known as a **duet**
- In forming compounds, atoms gain, lose, or share electrons to give a stable electron configuration characterized by 8 valence electrons, known as an **octet**

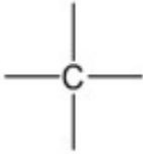
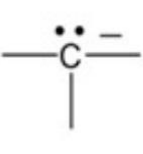
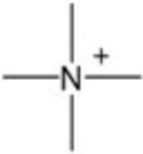
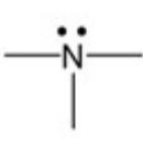
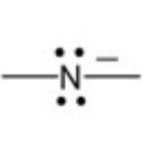
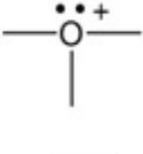
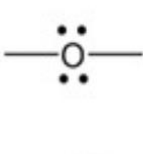
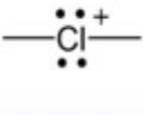


- The **octet rule** is the most useful in cases involving covalent bonds to second row elements **C**, **N**, **O**, and **F**
- Third row elements like **S**, **P** and **Cl** have additional atomic orbitals that can accommodate more than 8 electrons
 - Known as an **expanded octet**
- Details of bonding and orbitals will be discussed in Chapter 2

Drawing Lewis Structures

1. Write the Skeletal Structure
2. Count the total number of valence electrons
3. Attach atoms together with pairs of electrons
4. Complete octets, working **outside-in**
5. If all octets complete, give extra electrons to the central atom
6. If central atom does not have octet, bring in electrons from outside atoms to share

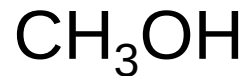
Lewis Structures: - Common Bonding Patterns

Atom	VE	+1	0	-1
C	4			
N	5			
O	6			
Halogen	7			

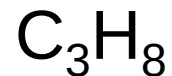
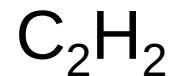
Lewis Structure Examples



Lewis Structure Examples-1

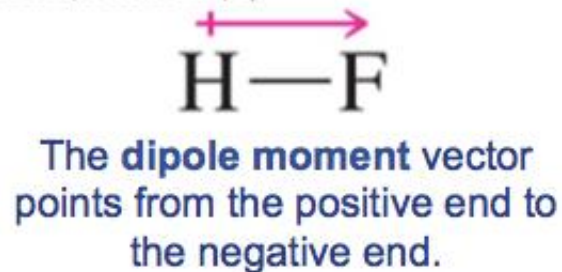
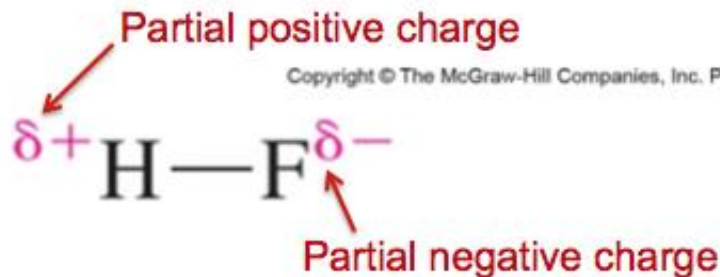


Lewis Structure Examples-2



Polar Covalent Bonds

- Two atoms of different electronegativity's share electrons unequally in a covalent bond. The result is a **polar covalent bond**
- The greater the difference in electronegativity, the more polarized the bond



Polar covalent bonds tend to be sites of reactivity in organic molecules.

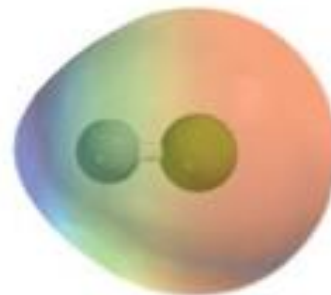
Periodic Trends in Electronegativity

- TABLE 1.3 Selected Values from the Pauling Electronegativity Scale

Period	Group number 1A	Group number 2A	Group number 3A	Group number 4A	Group number 5A	Group number 6A	Group number 7A
1	H 2.1						
2	Li 1.0	Be 1.5	B 2.0	C 2.5	N 3.0	O 3.5	F 4.0
3	Na 0.9	Mg 1.2	Al 1.5	Si 1.8	P 2.1	S 2.5	Cl 3.0
4	K 0.8	Ca 1.0					Br 2.8
5							I 2.5

Bond Polarity ($\Delta\chi$)

- Covalent - $\Delta\chi < 0.4$; e⁻ shared equally (H-H)
- Polar covalent - $\Delta\chi = 0.5 - 2.0$; bond dipole (H-F)
- Ionic - $\Delta\chi > 2.0$; e⁻ transferred (NaCl)



EWG

Dipole Moments

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TABLE 1.4 Selected Bond Dipole Moments			
Bond*	Dipole moment, D	Bond*	Dipole moment, D
H—F	1.7	C—F	1.4
H—Cl	1.1	C—O	0.7
H—Br	0.8	C—N	0.4
H—I	0.4	C=O	2.4
H—C	0.3	C=N	1.4
H—N	1.3	C≡N	3.6
H—O	1.5		

More polarized bonds are associated with greater differences in electronegativity and greater dipole moments.

Formal Charge

Formal charge = Valence electrons of the atom - Electron count

- Electron count = $\frac{1}{2}$ (# of shared electron) + # of unshared electron

Drawing Lewis Structures

1. Write the Skeletal Structure
2. Count the total number of valence electrons
3. Attach atoms together with pairs of electrons
4. Complete octets, working **outside-in**
5. If all octets complete, give extra electrons to the central atom
6. If central atom does not have octet, bring in electrons from outside atoms to share
7. **Calculate formal charge**

Incomplete Octets

[illegible]

lanthanum 57	cerium 58	praseodymium 59	neodymium 60	promethium 61	samarium 62	europium 63	gadolinium 64	terbium 65	dysprosium 66	holmium 67	erbium 68	thulium 69	ytterbium 70
La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb
138.91	140.12	140.91	144.24	[145]	150.36	151.96	157.25	158.93	162.50	164.93	167.26	168.93	173.04
actinium 89	thorium 90	protactinium 91	uranium 92	neptunium 93	plutonium 94	americium 95	curium 96	berkelium 97	californium 98	einsteinium 99	fermium 100	mendelevium 101	nobelium 102
Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No
[227]	232.04	231.04	238.03	[237]	[244]	[243]	[247]	[247]	[251]	[252]	[257]	[258]	[259]

Group 3 Examples



Expanded Octet

hydrogen 1 H 1.0079																	helium 2 He 4.0026						
lithium 3 Li 6.941	beryllium 4 Be 9.0122																	boron 5 B 10.811	carbon 6 C 12.011	nitrogen 7 N 14.007	oxygen 8 O 15.999	fluorine 9 F 18.998	neon 10 Ne 20.180
sodium 11 Na 22.990	magnesium 12 Mg 24.305																	aluminium 13 Al 26.982	silicon 14 Si 28.086	phosphorus 15 P 30.974	sulfur 16 S 32.065	chlorine 17 Cl 35.453	argon 18 Ar 39.948
potassium 19 K 39.098	calcium 20 Ca 40.078	scandium 21 Sc 44.956	titanium 22 Ti 47.867	vanadium 23 V 50.942	chromium 24 Cr 51.996	manganese 25 Mn 54.938	iron 26 Fe 55.845	cobalt 27 Co 58.933	nickel 28 Ni 58.693	copper 29 Cu 63.546	zinc 30 Zn 65.39	gallium 31 Ga 69.723	germanium 32 Ge 72.61	arsenic 33 As 74.922	selenium 34 Se 78.96	bromine 35 Br 79.904	krypton 36 Kr 83.80						
rubidium 37 Rb 85.468	strontium 38 Sr 87.62	yttrium 39 Y 88.906	zirconium 40 Zr 91.224	niobium 41 Nb 92.906	molybdenum 42 Mo 95.94	technetium 43 Tc [98]	ruthenium 44 Ru 101.07	rhodium 45 Rh 102.91	palladium 46 Pd 106.42	silver 47 Ag 107.87	cadmium 48 Cd 112.41	indium 49 In 114.82	tin 50 Sn 118.71	antimony 51 Sb 121.76	tellurium 52 Te 127.60	iodine 53 I 126.90	xenon 54 Xe 131.29						
caesium 55 Cs 132.91	barium 56 Ba 137.33	lutetium 71 Lu 174.97	hafnium 72 Hf 178.49	tantalum 73 Ta 180.95	tungsten 74 W 183.84	rhenium 75 Re 186.21	osmium 76 Os 190.23	iridium 77 Ir 192.22	platinum 78 Pt 195.08	gold 79 Au 196.97	mercury 80 Hg 200.59	thallium 81 Tl 204.38	lead 82 Pb 207.2	bismuth 83 Bi 208.98	polonium 84 Po [209]	astatine 85 At [210]	radon 86 Rn [222]						
francium 87 Fr [223]	radium 88 Ra [226]	lawrencium 103 Lr [262]	rutherfordium 104 Rf [261]	dubnium 105 Db [262]	seaborgium 106 Sg [266]	bohrium 107 Bh [264]	hassium 108 Hs [269]	meitnerium 109 Mt [268]	unnilium 110 Uun [271]	ununium 111 Uuu [272]	unbibium 112 Uub [277]	ununquadium 114 Uuq [289]											

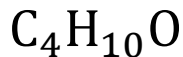
lanthanum 57 La 138.91	cerium 58 Ce 140.12	praseodymium 59 Pr 140.91	neodymium 60 Nd 144.24	promethium 61 Pm [145]	samarium 62 Sm 150.36	europium 63 Eu 151.96	gadolinium 64 Gd 157.25	terbium 65 Tb 158.93	dysprosium 66 Dy 162.50	holmium 67 Ho 164.93	erbium 68 Er 167.26	thulium 69 Tm 168.93	ytterbium 70 Yb 173.04
actinium 89 Ac [227]	thorium 90 Th 232.04	protactinium 91 Pa 231.04	uranium 92 U 238.03	neptunium 93 Np [237]	plutonium 94 Pu [244]	americium 95 Am [243]	curium 96 Cm [247]	berkelium 97 Bk [247]	californium 98 Cf [251]	einsteinium 99 Es [252]	fermium 100 Fm [257]	mendelevium 101 Md [258]	nobelium 102 No [259]

Expanded Octet Example

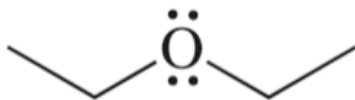


Levels of Organic Structure

- We think about the structures of organic molecules at various levels
 - The **composition** of a molecule is captured by its molecular formula



- The **constitution (connectivity)** of a molecule is captured by a structural formula showing how atoms are connected



- The **configuration** of a molecule is captured by a structural drawing showing the positions of atoms in space

Levels of Organic Structure

- We think about the structures of organic molecules at various levels
 - The **composition** of a molecule is captured by its molecular formula



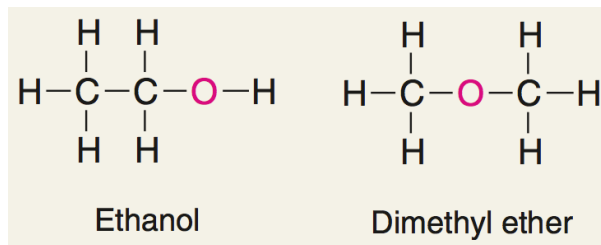
Compounds can have identical structures on a broad level but differ on a more specific level; such compounds are called **isomers**.



- The **configuration** of a molecule is captured by a structural drawing showing the positions of atoms in space

Constitutional Isomers

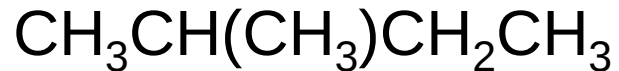
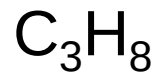
- **Constitutional isomers** have the same composition (molecular formula) but differ in how their atoms are connected



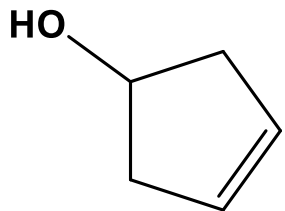
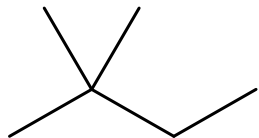
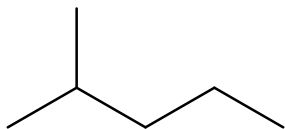
- Ethanol and dimethyl ether both have the molecular formula C_2H_6O , but they have different connectivity
- Constitutional isomers have different physical and chemical properties!
- Ethanol and Dimethyl ether have different physical properties.

WHY?

Line Structures



Line Structures



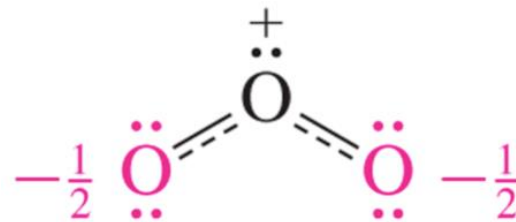
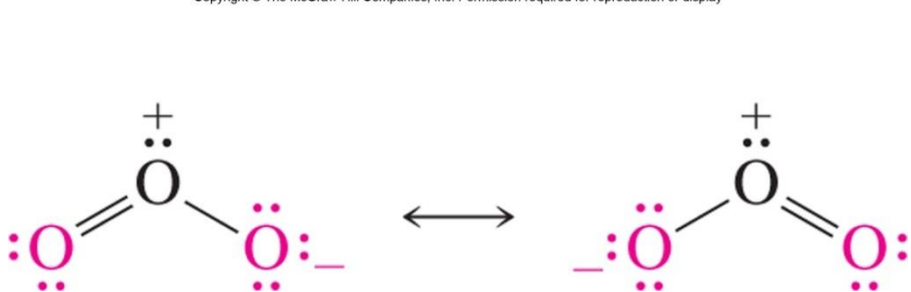
More Practice (on your own time)

- CO_2
- CH_3CF_3
- CH_3OH
- CH_3OCH_3
- $\text{CH}_3\text{CH}_2\text{OH}$
- $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$
- $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$
- $\text{CH}_3(\text{CH}_2)_2\text{CH}_3$
- $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_3$
- $\text{CH}(\text{CH}_3)_3$
- CH_2CH_2
- $\text{CH}_3\text{CHFCH}_3$
- $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$
- CH_3NHCH_3
- $\text{CH}_3\text{SCH}_2\text{CH}_3$
- CH_3COCH_3
- $\text{CH}_3\text{CO}_2\text{CH}_3$
- $\text{CH}_3\text{OCO}_2\text{CH}_3$
- CH_3CONH_2
- CH_3CN

Resonance

- Resonance represents the *delocalization* and *stabilization* of charge. Molecules with resonance are stabilized!

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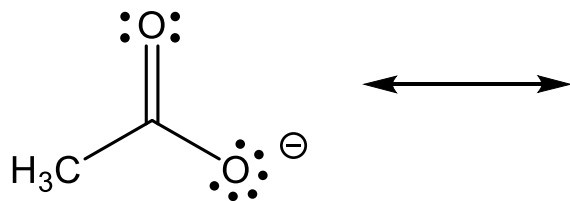


- The resonance hybrid

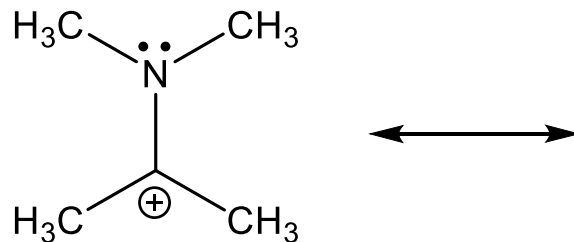
Rating Resonance Structures

- **N, O, F** will have filled octets
 - **C** occasionally will have only 6 electrons in some resonance structures
 - **B** will often have only 6 electrons
- Maximize number of covalent bonds
- Minimize formal charges
- Negative charge on most electronegative atom
- Minimize separation of formal charges

Resonance Structure Example



Resonance Structure Example



Which resonance structure is more stable?

Molecular Geometries

- Four equal atoms bonded to a central carbon
- Molecular Geometry Name:
- All bond angles equal to 109.5°
- **Examples**
 - Methane CH_4
 - Tetrafluoromethane CF_4
 - Tetrabromomethane CBr_4

Molecular Geometries

- Ammonia - NH_3
 - H-N-H angle = 107°
- Water - H_2O
 - H-O-H angle = 105°
- Boron trifluoride - BF_3
 - F-B-F angle = 120°
- Carbon dioxide - CO_2
 - O-C-O angle = 180°

Molecule Polarity

Acids and Bases: The Bronsted-Lowry View

- Brønsted-Lowry definition:
 - An acid is a proton donor
 - A base is a proton acceptor
- Proton Transfer from HBr to Water

Acid Strength

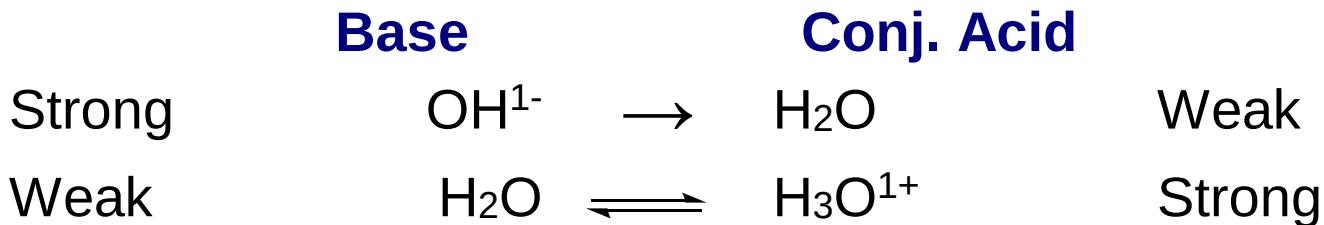
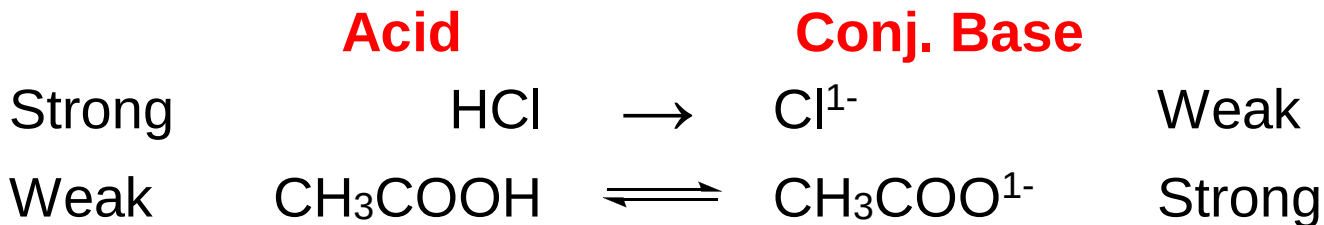
- $\text{H-X} + \text{H}_2\text{O} \rightleftharpoons \text{H}_3\text{O}^+ + \text{X}^-$
- Acid strength is measured by the dissociation constant K_a

Factors Effecting Acidity

- Delocalization or resonance stabilization
- Bond strength
 - Higher strength of the acid—H bond gives lower acidity
- Electronegativity
 - Within a row of the periodic table, higher electronegativity gives higher acidity

Acid/Base Strength General Rules

The strength of an acid/base is dependent on the stability of its conjugate form

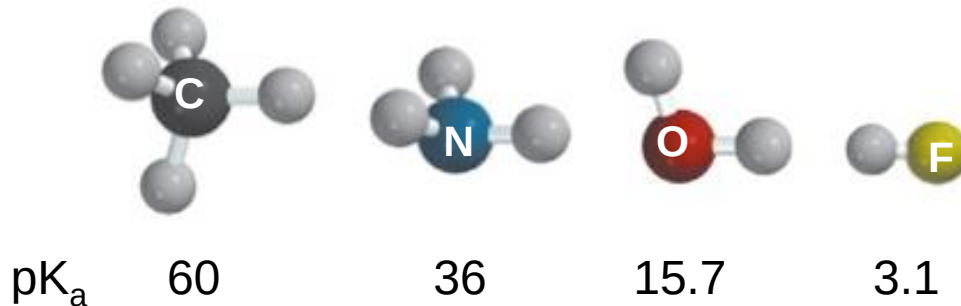


Acid Strength General Rules

- Size of atom – bond strength decreases going down a group; stronger acid

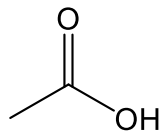


- Electronegativity – as χ increases, bond becomes more polarized; stronger acid

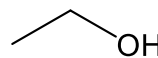
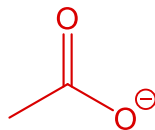


Acid Strength General Rules

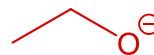
- Resonance in conjugate base



$pK_a = 4.7$



$pK_a = 16$



- Inductive effects – EWG near H increases acidity



CH_3CH_2O-H
Ethanol
 pK_a 16

CH_3CH_2OH
 pK_a 16



CF_3CH_2O-H
2,2,2-Trifluoroethanol
11.3

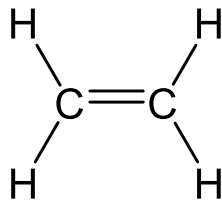
CF_3CH_2OH
11.3

Acid Strength General Rules

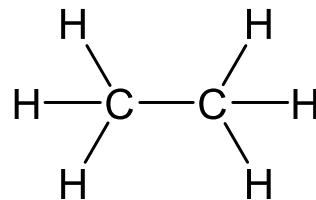
- Hybridization – if an atom bearing a negative charge has more s-character, the anion will be more stable



$\text{pK}_a = 26$



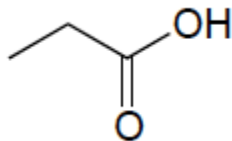
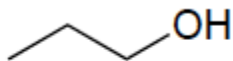
45



62

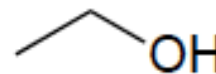
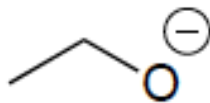
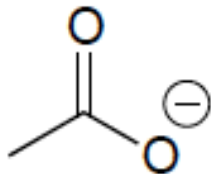
Strength of Acid Examples

- Rank the acidity for the following structures from most acidic to least acidic.



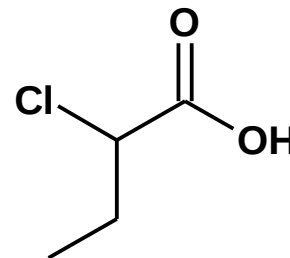
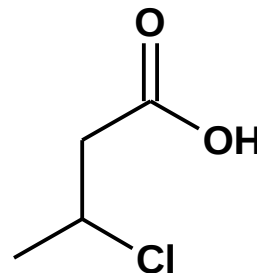
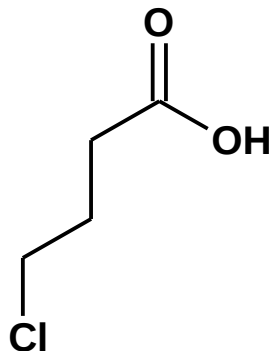
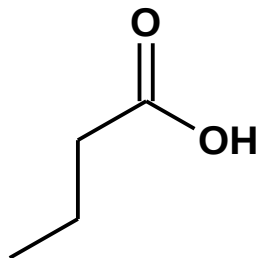
Strength of Acid Examples

- Rank the basicity for the following structures from least basic to most basic. (remember, most basic will be the least stable base and form the weakest conjugate acid)



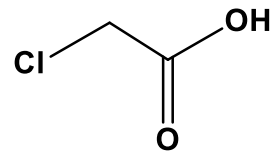
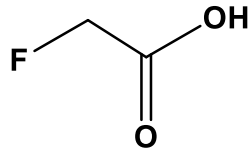
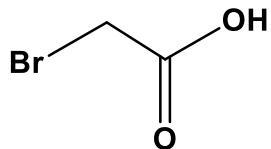
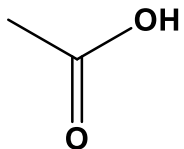
Strength of Acid Examples

- Rank the acidity for the following structures from least acidic to most acidic.



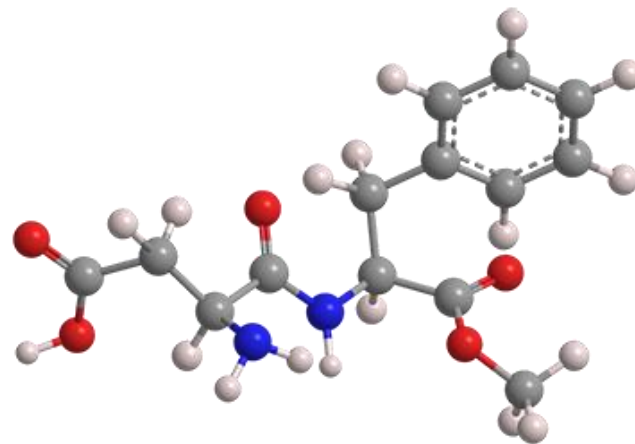
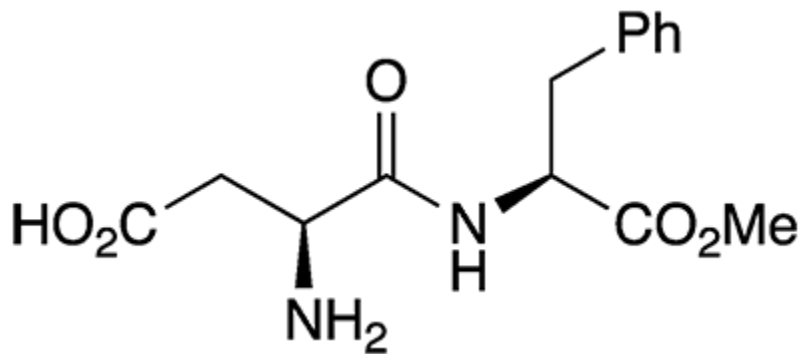
Strength of Acid Examples

- Rank the acidity for the following structures from least acidic to most acidic.



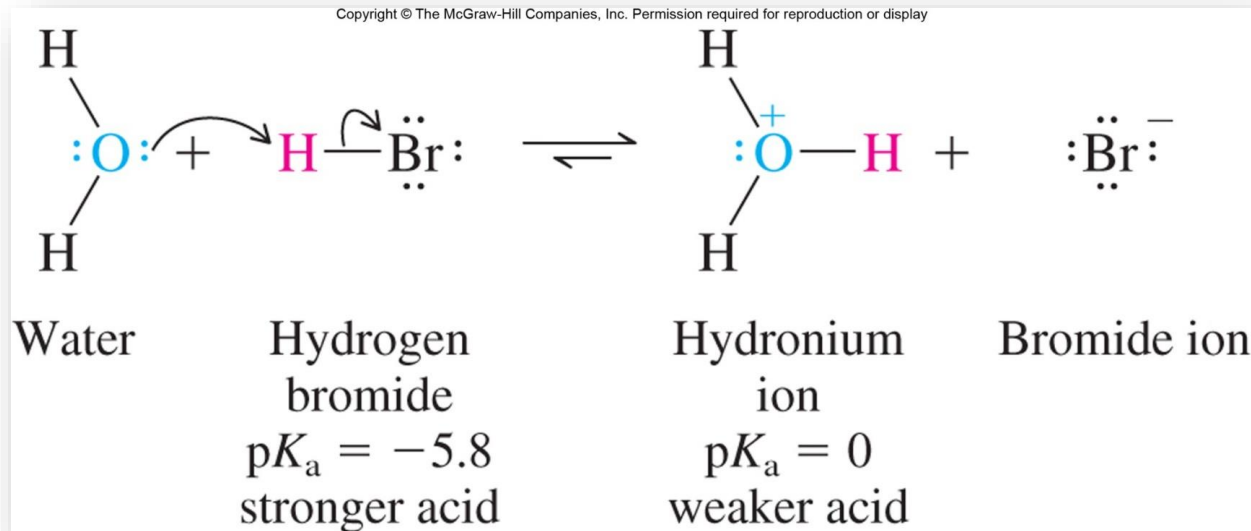
Identify the Most Acidic Proton

- The structure of aspartame, an artificial sweetener is shown. Rank the acidity of the protons from 1-3, 1 being the strongest.

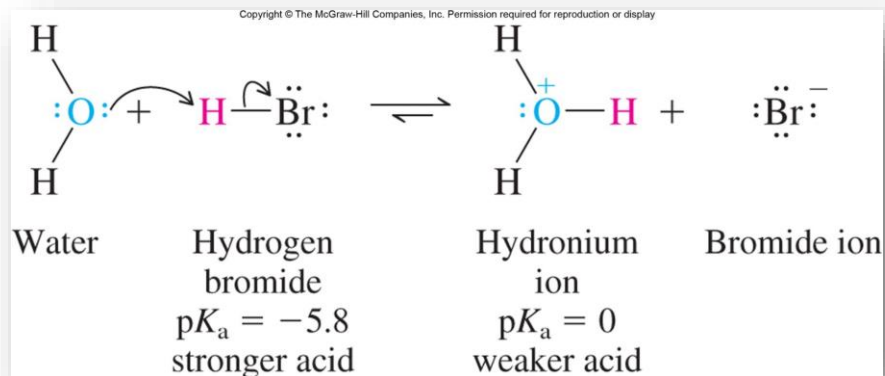


Acid-Base Equilibria

- In general, an acid-base reaction is favorable when the stronger acid is on the left and the weaker acid is on the right.



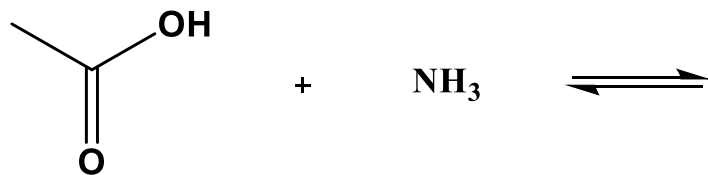
Acid-Base Equilibria Calculation



$$K_{\text{eq}} = \frac{K_a \text{ of reactant acid}}{K_a \text{ of product acid}}$$

Strength of Acid Examples

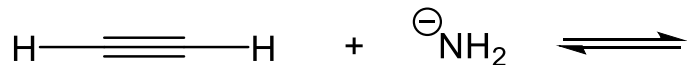
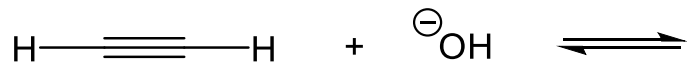
- Complete the following acid-base reaction between acetic acid and ammonia.



- The pK_a of acetic acid is 4.7 and pK_a of the ammonium ion is 9.3. Calculate K_{eq} . Will the reaction favor the reactants or products?

Strength of Acid Examples

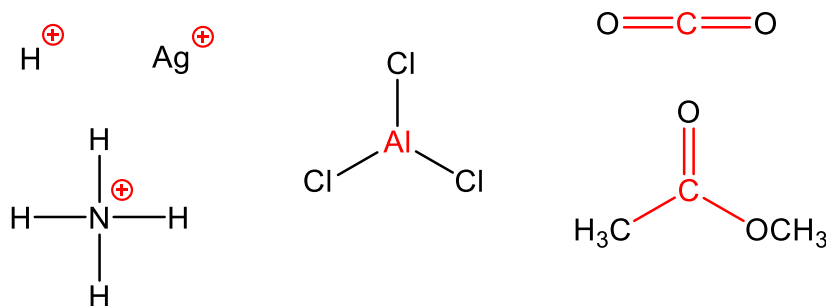
- Complete the following reactions. Given that ethyne has a pK_a value of 25, water has a pK_a value of 15.7, and ammonia has a pK_a value of 36, determine whether the reactants or products are favored for each reaction.



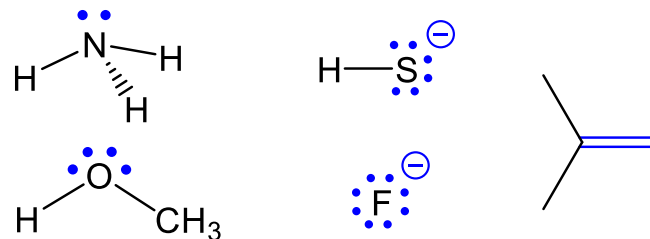
- Which is a more effective base to deprotonate ethyne?

Lewis Acids and Bases

- Lewis Acid: a molecule that can accept a pair of electrons to form a new bond



- Lewis Base: a molecule that can donate a pair of electrons to form a new bond



Lewis Acids and Bases

- Lewis Acid: a molecule that can accept a pair of electrons to form a new bond

ELECTROPHILE

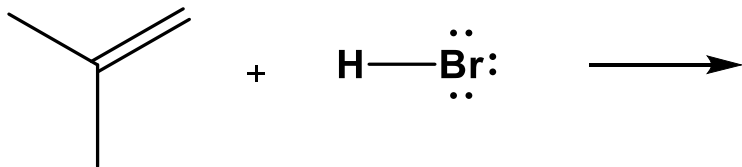
- Lewis Base: a molecule that can donate a pair of electrons to form a new bond

NUCLEOPHILE

Arrow Pushing

Lewis Acid/Base Example

- Complete the Lewis acid/base reaction below using the curved-arrow notation to show the steps of the reaction.



Lewis Acid/Base Example

- Write the Lewis acid/base reaction between water and carbon dioxide.

TABLE 1.8 Acidity Constants (pK_a) of Acids				
Acid	pK_a	Formula	Conjugate base	Discussed in section
Hydrogen iodide	-10.4	$\text{HI}:$	$:\ddot{\text{I}}:^-$	1.13; 1.14
Hydrogen bromide	-5.8	$\text{H}\ddot{\text{B}}\text{r}:$	$:\ddot{\text{Br}}:^-$	1.13; 1.14
Sulfuric acid	-4.8	$\text{H}\ddot{\text{O}}\text{SO}_3\text{H}$	$:\ddot{\text{O}}\text{SO}_3\text{H}^-$	1.14
Hydrogen chloride	-3.9	$\text{H}\ddot{\text{Cl}}:$	$:\ddot{\text{Cl}}:^-$	1.13; 1.14
Hydronium ion*	-1.7	$\text{H}_3\ddot{\text{O}}:$	$\text{H}_2\ddot{\text{O}}^-$	1.14
Nitric acid	-1.4	$\text{H}\ddot{\text{O}}\text{NO}_2$	$:\ddot{\text{O}}\text{NO}_2^-$	1.13
Hydrogen sulfate ion	2.0	$:\ddot{\text{O}}\text{SO}_3\text{H}$	$:\ddot{\text{O}}\text{SO}_3^-$	1.14
Hydrogen fluoride	3.1	$\text{H}\ddot{\text{F}}:$	$:\ddot{\text{F}}:^-$	1.13
Anilinium ion	4.6	$\text{C}_6\text{H}_5\text{NH}_3^+$	$\text{C}_6\text{H}_5\text{NH}_2$	22.4
Acetic acid	4.7	CH_3COOH	CH_3COO^-	1.13; 1.14; 19.5
Pyridinium ion	5.2	$\text{C}_5\text{H}_5\text{NH}^+$	$\text{C}_5\text{H}_5\text{N}$	12.22; 22.4
Carbonic acid	6.4	H_2CO_3	$:\ddot{\text{O}}\text{CO}_2\text{H}^-$	19.9
Hydrogen sulfide	7.0	$\text{H}_2\ddot{\text{S}}$	HS^-	7.19
2,4-Pentanedione	9	$\text{CH}_3\text{COCH}_2\text{COCH}_3$	$\text{CH}_3\text{COCH}^-\text{COCH}_3$	21.2
Hydrogen cyanide	9.1	$\text{HC}\equiv\text{N}:$	$:\text{C}\equiv\text{N}^-$	7.19
Ammonium ion	9.3	$^+\text{NH}_4$	$:\text{NH}_3$	22.4
Glycine	9.6	$\text{H}_3\text{N}^+\text{CH}_2\text{COO}^-$	$\text{H}_2\text{NCH}_2\text{COO}^-$	26.3
Phenol	10	$\text{C}_6\text{H}_5\text{OH}$	$\text{C}_6\text{H}_5\text{O}^-$	1.14; 23.4
Hydrogen carbonate ion	10.2	$\text{H}\ddot{\text{O}}\text{CO}_2^-$	$:\ddot{\text{O}}\text{CO}_2^-$	19.9
Methanethiol	10.7	$\text{CH}_3\ddot{\text{S}}\text{H}$	$\text{CH}_3\ddot{\text{S}}^-$	16.12
Dimethylammonium ion	10.7	$(\text{CH}_3)_2\text{NH}_2^+$	$(\text{CH}_3)_2\text{NH}$	22.4

continued

*For acid-base reactions in which water is the solvent, the pK_a of H_3O^+ is zero and the pK_a of H_2O is 14.

TABLE 1.8 Acidity Constants (pK_a) of Acids (Continued)				
Acid	pK_a	Formula	Conjugate base	Discussed in section
Ethyl acetoacetate	11	$\text{CH}_3\text{COCH}_2\text{COOCH}_2\text{CH}_3$	$\text{CH}_3\text{COCH}^-\text{COOCH}_2\text{CH}_3$	21.2
Piperidinium ion	11.2	$\text{C}_5\text{H}_{11}\text{NH}^+$	$\text{C}_5\text{H}_{11}\text{N-H}$	22.4
Diethyl malonate	13	$\text{CH}_3\text{CH}_2\text{COOCH}_2\text{CH}_3$	$\text{CH}_3\text{CH}^-\text{COOCH}_2\text{CH}_3$	21.2
Methanol	15.2	$\text{CH}_3\ddot{\text{O}}\text{H}$	$\text{CH}_3\ddot{\text{O}}^-$	1.13; 15.4
2-Methylpropanal	15.5	$(\text{CH}_3)_3\text{CCHO}$	$(\text{CH}_3)_3\text{CCH}^-$	21.2
Water*	15.7	$\text{H}_2\ddot{\text{O}}:$	HO^-	1.13
Ethanol	16	$\text{CH}_3\text{CH}_2\ddot{\text{O}}\text{H}$	$\text{CH}_3\text{CH}_2\ddot{\text{O}}^-$	1.13
Cyclopentadiene	16	C_5H_6	C_5H_5^-	12.20
Isopropyl alcohol	17	$(\text{CH}_3)_2\text{CHOH}$	$(\text{CH}_3)_2\text{CHO}^-$	1.13
tert-Butyl alcohol	18	$(\text{CH}_3)_3\text{COH}$	$(\text{CH}_3)_3\text{CO}^-$	1.13
Acetone	19	CH_3COCH_3	$\text{CH}_3\text{C}^-\text{CH}_3$	21.2
Ethyl acetate	25.6	$\text{CH}_3\text{COOCH}_2\text{CH}_3$	$\text{CH}_3\text{C}^-\text{OCH}_2\text{CH}_3$	21.2
Acetylene	26	$\text{HC}\equiv\text{CH}$	$:\text{C}\equiv\text{CH}^-$	9.5
Hydrogen	35	H-H	H^-	21.8
Ammonia	36	$:\text{NH}_3$	$\text{H}_2\ddot{\text{N}}^-$	1.13; 1.14; 15.4
Diisopropylamine	36	$(\text{CH}_3)_2\text{CHNHCH}_3$	$(\text{CH}_3)_2\text{CHN}^-\text{CH}_3$	21.2

*For acid-base reactions in which water is the solvent, the pK_a of H_3O^+ is zero and the pK_a of H_2O is 14.